

Medical image analysis: Refresher

Just pointers and reminders

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Outline

- Basic notation
- Probabilistic modeling
- Optimization, cost function and regularization
- Linear basis models and function parameterizations
- Spatial transformations
- Derivative approximations

Basic notation

■ Continuous version

A volumetric grayscale image is a function

$I : \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^3$ is the image domain

$I(x)$ is the intensity at point $x \in \Omega$, $x = [x_1, x_2, x_3]$

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- For dynamic images with temporal information

$I : \Omega \times \mathbb{R}^+ \rightarrow \mathbb{R}^N$

$I(x, t)$ is the intensity at point x and time t

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- Probabilistic modeling
 - PDF, CDF, PMF
 - Histogram of an image
 - Conditionals and the Bayes' Rule
 - Posterior distribution, MAP and MLE
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PDF, CDF, PMF

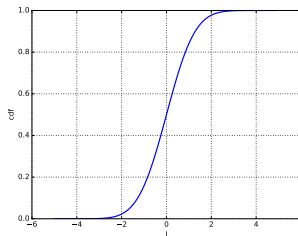
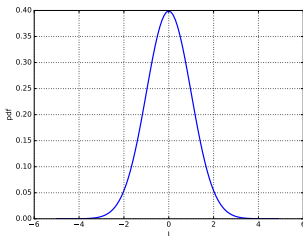
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 $p(i)$ as its *probability density function (PDF)*
 $P(i) = \Pr[I \leq i] = \int_{-\infty}^i p(j) dj$ as its *cumulative distribution function (CDF)*

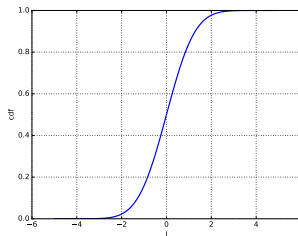
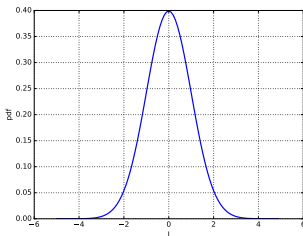
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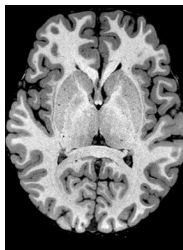
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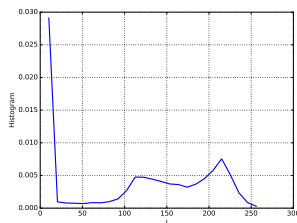


- For discrete random variables, L , we define
 $p(l) = p(L = l)$ as its *probability mass function (PMF)*
PMF can be seen as a PDF is often named as PDF in scientific articles

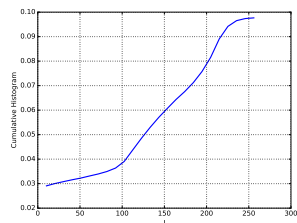
Histogram of an image



Image



Histogram



Cumulative histogram

If we consider each pixel intensity as an independent realization of the random variable I then the histogram is an approximation to the PDF and cumulative histogram is an approximation to the CDF

Conditionals and the Bayes' Rule

- When we have two variables such as I and L

$p(i, l)$: joint distribution

$p(i) = \sum_{l=0}^N p(i, l)$ and $p(l) = \int_{-\infty}^{\infty} p(i, l) di$: marginal distributions

$p(i|l)$ and $p(l|i)$: conditional distributions

$p(i, l) = p(i|l)p(l) = p(l|i)p(i)$

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- In a large variety of problems one of them is observed and the other not. Assume i is observed in that case

$p(i|l)$: likelihood

$p(l)$: prior distribution

$p(l|i)$: posterior distribution

Posterior distribution, MAP and MLE

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- Example problems: image enhancement, segmentation and even registration.

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- MLE is the same as MAP when the prior is uniform, i.e. $p(I) = c, \forall I$

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Data term and regularization

- Most problems in medical image analysis are formulated as optimization problems

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- Image registration: Determine T between images I and J

$$\arg_T \min \int (I(x) - J(T(x)))^2 dx + \int \|\alpha \Delta T(x) + \beta \nabla(\nabla \cdot T(x)) - \gamma T(x)\|_2^2 dx$$

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- The MAP estimate is also in the same form

$$\arg_{\theta} \max \log p(i|\theta) + \log p(\theta) = \arg_{\theta} \min -\log p(i|\theta) - \log p(\theta)$$

Regularizers can be thought of as priors with $-\log p(\theta) \propto \mathcal{R}(\theta)$

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- Some references

- Convex Optimization, Boyd and Vandenberghe, Cambridge University Press, 2004.
 - Nonlinear Programming, Bertsekas, Athena Scientific, 2016.

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 - Basics
 - Function parameterizations with linear basis models
 - Kernel-based parameterization
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- Function parameterization for optimization, few parameters to describe the function
- In finite vector spaces

$$\vec{v} = a_1 \vec{b}_1 + a_2 \vec{b}_2 + \dots + a_d \vec{b}_d = \mathbf{B} \vec{a}$$

$\vec{b}_i$ are the basis functions and a_i coefficients.

if $\vec{b}_i$ are orthogonal, i.e. $\vec{b}_i^T \vec{b}_j = 0, \forall i \neq j$ then $a_i = \vec{v}^T \vec{b}_i / \|\vec{b}_i\|_2$.

if not then Ordinary Least Square (OLS) regression must be performed

$$\arg_{\vec{a}} \min \|\vec{v} - \mathbf{B} \vec{a}\|_2^2 = \left(\mathbf{B}^T \mathbf{B} \right)^{-1} \mathbf{B}^T \vec{v}$$

For known $\vec{b}_i$, $\vec{a}$ can be a parameterization for $\vec{v}$

Function parameterizations with linear basis models

Global parameterizations

$$f(x) = \sum_i^d a_i b_i(x)$$

where $b_i(x)$ are smooth basis functions.

- The parameterization is $\vec{a}$
- To determine $\vec{a}$, the space is discretized and a $\mathbf{B}$ matrix is formed.
- Note that this parameterization cannot represent all functions.
- Examples:
 - polynomials: bias-field correction in MRI
 - splines: non-linear registration

Kernel-based parameterization

$$f(x) = \sum_{i=1}^d a_i K(x, x_i)$$

where x_i are the control points and $K(x, x_i)$ is a kernel function

- Radial basis functions are often used: $K(x, x_i) = K(\|x - x_i\|_2)$
- Popular radial basis functions:
Gaussian

$$K(\|x - x_i\|_2) = \exp - \frac{\|x - x_i\|_2^2}{\sigma^2}$$

Thin-plate spline

$$K(\|x - x_i\|_2) = \|x - x_i\|_2^2 \ln(\|x - x_i\|_2)$$

- Example: landmark based registration, linear/non-linear registration, kernel density estimation

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 - 6 dof in 3D - 3 translation and 3 rotation
 - Intra-subject registration, multi-modal intra-subject registration

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 - Intra-subject registration, multi-modal intra-subject registration
- Similarity transformation
 - 4 dof in 2D - Rigid + scale
 - 7 dof in 3D - Rigid + scale
 - Used for coarse alignment in inter-subject. Initialization for non-linear

Linear transformations

$$T(\vec{x}) = \mathbf{T}\vec{x},$$

- Used in linear image registration

- Rigid

- 3 dof in 2D - 2 translation and 1 rotation
- 6 dof in 3D - 3 translation and 3 rotation
- Intra-subject registration, multi-modal intra-subject registration

- Similarity transformation

- 4 dof in 2D - Rigid + scale
- 7 dof in 3D - Rigid + scale
- Used for coarse alignment in inter-subject. Initialization for non-linear

- Affine transformation

- 6 dof in 2D
- 12 dof in 3D
- Often not used in full dof
- Instead, 9 dof transformation (3 translation + 3 rotation + 3 scale) may be preferred.

$$\mathbf{T} = \underbrace{\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}}_{\text{scale}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}}_{\text{rotation around x-axis}} \underbrace{\begin{bmatrix} \cos(\phi) & 0 & \sin(\phi) \\ 0 & 1 & 0 \\ -\sin(\phi) & 0 & \cos(\phi) \end{bmatrix}}_{\text{rotation around y-axis}} \underbrace{\begin{bmatrix} \cos(\gamma) & \sin(\gamma) & 0 \\ -\sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{rotation around z-axis}} + \underbrace{\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}}_{\text{translation}}$$

Non-linear transformations

$$T(\vec{x}) = \vec{x} + \underbrace{\vec{u}(\vec{x})}_{\text{displacement field}}$$

- Used in non-linear registration.
- Different models are used.
- Linear basis function models - additive
- More advanced techniques based on composition of transformations

$$T(\vec{x}) = T_n \circ T_{n-1} \circ \cdots \circ T_1(\vec{x})$$

where $\circ$ is basic function composition

Allows modeling diffeomorphisms

see Computational Anatomy

Transformation related identities

$$T(\vec{x}) = [T_1(\vec{x}), T_2(\vec{x}), T_3(\vec{x})]^T$$

- Jacobian

$$\mathbf{J} = \begin{bmatrix} \frac{\partial T_1}{\partial x_1} & \frac{\partial T_1}{\partial x_2} & \frac{\partial T_1}{\partial x_3} \\ \frac{\partial T_2}{\partial x_1} & \frac{\partial T_2}{\partial x_2} & \frac{\partial T_2}{\partial x_3} \\ \frac{\partial T_3}{\partial x_1} & \frac{\partial T_3}{\partial x_2} & \frac{\partial T_3}{\partial x_3} \end{bmatrix}$$

Jacobian determinant quantifies local volumetric change.

$\det(\mathbf{J}) = 1$: no change, $\det(\mathbf{J}) < 1$: compression, $\det(\mathbf{J}) > 1$: expansion

- Divergence: $\nabla \cdot T$ - trace of the Jacobian.

Gives information about the amount of compression and expansion as well.

- Curl - $\nabla \times T$ - information on the amount of infinitesimal rotation

Outline

- Basic notation
- Probabilistic modeling
- Optimization, cost function and regularization
- Linear basis models and function parameterizations
- Spatial transformations
- Derivative approximations

Derivative approximations

- Finite difference approximations is used the most often
- First order derivative at a grid point x_0 with grid spacing Δx

$$\begin{aligned}\frac{df}{dx}|_{x_0} &\approx \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \\ &\approx \frac{f(x_0) - f(x_0 - \Delta x)}{\Delta x} \\ &\approx \frac{f(x_0 + \Delta x) - f(x_0 - \Delta x)}{2\Delta x}\end{aligned}$$

- Second order derivative at a grid point x_0 with grid spacing Δx

$$\frac{d^2f}{dx^2}|_{x_0} \approx \frac{f(x_0 + \Delta x) - 2f(x_0) + f(x_0 - \Delta x)}{\Delta x^2}$$

- Many different approximations exist